# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2– Pseudocode Writing Task

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|---|---|---|---|
| **Course Code:** | **ITA05** | **Course Title:** | **Computer Vision** |
| **CO Assessed:** | **CO5 – Naturalization (independent application using OpenCV)P5** | **Assessment:** | **Assessment Tool 2 – Pseudocode Writing Task** |
| **Weightage:** | **25% of CIA** | **Total Marks:** | **100** |
| **CO4 Statement:** | *Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos. (BTL6)* | | |
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| **Section / Batch:** | B. Tech AI & ML | **Date:** | 26 - 08 - 2026 |

**Code of Conduct**
*I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*
Signature of the Student: *S. Susannal Devapriyam*

| **Criteria** | **Weigh tage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** | **Marks** |
|---|---|---|---|---|---|---|
| Problem Understanding | 20% | Identifies all inputs, outputs, constraints accurately | Minor missing elements | Partial understanding | Incorrect interpretation | |
| Algorithm Design | 30% | Complete, correct, and logically sound solution | Minor logical gaps | Partially correct logic | Incorrect approach | |
| Use of Control Structures | 20% | Efficient and appropriate use of loops, conditions, functions/modules | Minor inefficiencies | Limited or basic usage | Incorrect or missing constructs | |
| Pseudocode Structure | 10% | Well-structured, clear, consistent format, easy to follow | Mostly clear with minor issues | Difficult to follow in parts | Poorly structured and unclear | |
| Completeness | 20% | Handles all cases; optimized approach | Handles most cases; reasonable efficiency | Handles basic cases only | Incomplete or inefficient | |

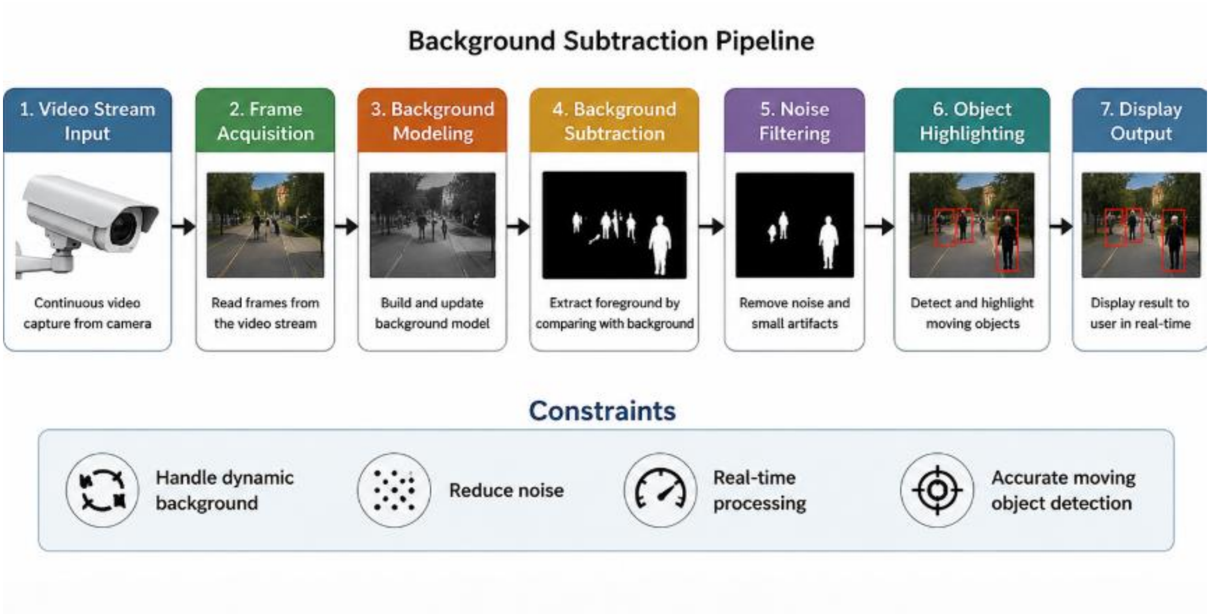
# Pseudocode Writing Task

## Background Subtraction Pipeline :

### 1. Problem Understanding

A surveillance system continuously captures video frames and must identify moving objects. The system should separate foreground objects from the background, reduce unwanted noise, handle dynamic background changes, and display the detected moving objects.

**Basic Pipeline Diagram:**




### 2. Proposed Approach

The system first reads frames from the video stream and creates a background model. Each new frame is compared with the background to identify differences. The resulting foreground mask is filtered to remove noise. Finally, significant moving regions are detected and highlighted.

### 3. Pseudocode

BEGIN

    OPEN video stream

```
INITIALIZE background model as NULL

SET threshold value
SET minimum object area

WHILE video stream is active DO

   READ current frame

   IF frame is not available THEN
      BREAK
   END IF

   CONVERT frame to grayscale

   APPLY Gaussian Blur to reduce noise

   IF background model is NULL THEN
      SET current frame as background model
      CONTINUE
   END IF

   COMPUTE difference between
   current frame and background model

   APPLY threshold to create foreground mask

   APPLY morphological opening
   to remove small noise

   APPLY morphological closing
   to connect object regions

   FIND contours in foreground mask

   FOR each contour DO

      COMPUTE contour area

      IF contour area > minimum object area THEN

         DRAW bounding box around object

         LABEL it as "Moving Object"

      END IF

   END FOR

   UPDATE background model gradually
```

```
        DISPLAY processed frame

        IF user presses 'Q' THEN
            BREAK
        END IF

    END WHILE

    RELEASE video stream

END
```

**4. Explanation of Major Steps**

**1. Video Frame Acquisition**

The system continuously reads frames from the surveillance camera or recorded video source.

**2. Background Modeling**

A background model represents the static part of the surveillance scene for comparison.

**3. Background Subtraction**

The current frame is compared with the background to identify significant changes.

**4. Thresholding**

The difference image is converted into a binary mask containing foreground and background regions.

**5. Noise Filtering**

Gaussian filtering and morphological operations remove unwanted pixels and improve object boundaries.

**6. Object Detection**

Contours are identified from the filtered foreground mask to locate moving objects.

### 7. Object Highlighting

Bounding boxes are drawn around significant objects to clearly indicate detected movement.

### 5. Handling Dynamic Background

Dynamic backgrounds may include moving leaves, rain, shadows, or gradual lighting changes.

The system handles this using adaptive background updating**.**

Old Background

+

Current Stable Frame

↓

Updated Background

This gradual updating helps the system adapt without treating permanent environmental changes as moving objects.

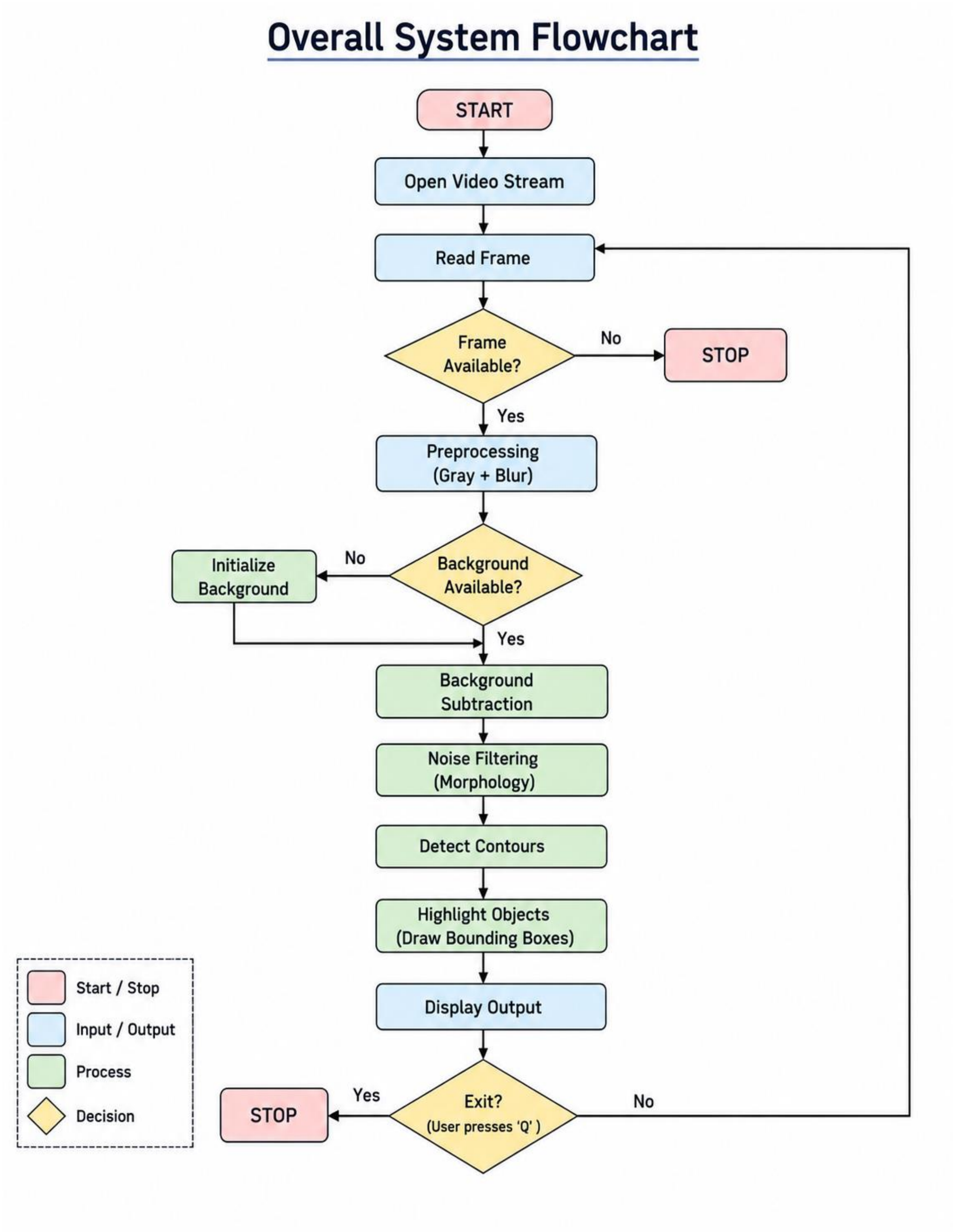
### 6. Noise Reduction Process

Noise may produce unwanted small regions in the foreground mask.

Foreground Mask
↓
Gaussian Filtering
↓
Morphological Opening
↓
Remove Small Noise
↓
Morphological Closing
↓
Clean Object Regions

- **Gaussian Blur** reduces sensor and high-frequency noise.
- **Opening** removes isolated small pixels.
- **Closing** fills small gaps inside detected objects.

## 7. Overall System Flowchart



## 8. Conclusion

The proposed background subtraction pipeline effectively detects moving objects from a surveillance video. Adaptive background modeling handles dynamic environmental changes, while filtering techniques reduce noise. Contour detection and bounding boxes finally identify and highlight significant moving objects.

**Why this approach satisfies the constraints**

- Reads continuous video streams efficiently.
- Separates foreground objects from static backgrounds.
- Handles dynamic backgrounds through adaptive updating.
- Reduces noise using filtering and morphology.
- Highlights detected moving objects clearly.